



Investigating Eclipsing Binaries and Their Eccentric Behaviours

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Abstract

The study aimed to identify and characterise detached eclipsing binaries that could serve as benchmark systems for the upcoming PLATO mission. Using TESS photometry and GAIA DR3 data, ten systems were analysed through phase-folding light curve modelling and creating a final colour-magnitude diagram of all ten binaries. JKTEBOP was used to derive key system parameters, including inclination, fractional radii, and, where necessary, eccentricity terms. There were some challenges, such as missing GAIA data for TIC 307084982, but the TESS data were overlaid over the model fit and showed behaviour of eccentricity. Other systems like TIC 7695666 and TIC 78568736 also showed eccentric behaviour, aligning with literature that binaries with large periods are likely to be eccentric. In the end, six systems met the PLATO p1 sample specification, and all ten met the P5 criteria. Results show that clear and careful modelling can deliver a high-quality benchmark for PLATO.

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1 Introduction

1.1 PLATO's mission

The European Space Agency (ESA) developed a new satellite, PLATO (PLANetary Transits and Oscillations of stars), scheduled to launch in 2026. PLATO's aim and mission are to find and study exoplanets, simultaneously advancing our understanding of stellar physics, which will be a significant leap for modern astronomy. Over 4 years, PLATO's main mission is to discover habitable planets, favouring Earth-like planets in the habitable zones of their host stars. Using our solar system as the benchmark, PLATO will focus on finding solar-type stars to find Earth-like planets (Rauer et al. 2024). To achieve this mission, PLATO must conduct complex asteroseismic studies, using current understanding of planetary science and stellar astrophysics.

Exoplanets and their host stars can only be studied and considered suitable or unsuitable for PLATO by examining the host star's specific characteristics first and observing the planet, if it is a solar-type star. Assessing the stellar parameters is key; looking at the visual magnitudes, stellar radius, and stellar mass is important. However, there are errors when observing and calculating these parameters, where the large errors of stellar radius can influence the increase in error of the derived planetary radius. This means minimising errors is important so we can determine an accurate depiction of exoplanets, their composition and atmospheric retention. PLATO will use its high precision photometry to give an accurate visual magnitude, minimising the errors of stellar parameters (Catala et al. 2010). Adding a sample of benchmark eclipsing binaries (EB), which are known independently, will help PLATO derive masses, radii and orbital parameters to high accuracy (Lebreton & Goupil 2014). Our study helps in this as we analysed a selection of stars that could potentially be benchmarks.

PLATO will focus on the long Observation Phase South 2 (LOPS2) field in the southern sky. LOPS2 field was chosen due to several important factors: NASA's TESS mission has provided photometric data and candidates for PLATO to observe (Nascimbeni 2021). Another important factor is the high density of bright solar-like stars, offering PLATO large data for filtering through and finding habitable planets. Some solar-like stars are suspected of hosting habitable planets, so PLATO may be able to confirm and characterise them. The southern sky has been studied extensively by telescopes such as the Very Large Telescope (VLT) and the Southern African Large Telescope (SALT), enabling follow-up observations of PLATO targets.

Stars in EBs are selected and considered good candidates due to their apparent visual magnitudes. Their magnitudes are then categorised into several groups. The groups are

$P1$, $P2$, $P4$ and $P5$, but the mission is focused on $P1$ and $P5$ sample specifications, as they are main sequence stars, and therefore of high importance (Rauer et al. 2024). The $P1$ samples include bright stars within the main-sequence, with a spectral range between $F5 - K7$ and a visual magnitude of $V \leq 11$ mag. These $P1$ stars are ideal for analysis as PLATO’s high photometric accuracy can detect detailed asteroseismic signals. While PLATO will favour $P1$ stars, the $P5$ sample is still of high importance. These stars are cooler and dimmer main sequence stars, including late K-type stars and a visual magnitude of $V \leq 13$ mag. Although they are fainter, they will contribute by providing a diverse dataset and helping astronomers understand the evolution and diversity of planets and stars (Rauer et al. 2024).

Despite PLATO’s high photometric accuracy, benchmark EBs are essential for reliable measurements. Finding these benchmarks involves filtering through thousands of EBs, looking at their light curves and modelling candidates and from these obtained parameters. The project aimed to do this by compiling, analysing and validating high-priority EBs. The selected EBs within this project are crucial in improving model precision and creating accurate stellar properties that PLATO can use.

1.2 Theoretical Background

Eclipsing Binaries (EBs) are binary systems whose brightness changes periodically when their two components pass each other. EBs are identified and analysed using light curves, from the observer’s line of sight, astronomers can identify which is the larger and brighter star in a binary system. Astronomers use light curves to identify: primary and secondary depths, the shape of the eclipses (allowing observers to see the type of interaction between the stars), stellar radii, orbital inclination, orbital period and eccentricity. With these parameters, astronomers can derive the stellar masses, essential to determine if the EBs analysed are solar-like (Southworth 2013). Although lone star systems require model-based assumptions to estimate their parameters, EBs have a model-independent approach to stellar characterisation, making them easier to study (Aerts et al. 2019).

Asteroseismology uses oscillation models to estimate stellar properties; however, this method involves external calibration due to the empirical scaling relation. To improve the reliability of the seismic inferences across all the PLATO samples, deriving properties of the individual detached EBs and then comparing with derived EB parameters (Maxted et al. 2022). This makes detached binaries favoured benchmarks as they have spherical symmetry not influenced by the other stars due to the Roche lobe. Leading to a clean and simple light curve models, used to analyse the parameters.

Although detached binaries provide us with an ideal benchmark, there are other factors

to consider when ensuring the reliability of the data received. Third light and stellar spots are two examples, and these risks are reduced using the morphology index as a filtering tool. EBs which were reliable were found to have a morphology of < 0.6 and, therefore, a priority for modelling (Prša et al. 2022).

Once the candidate systems were selected, their full characterisation involved combining data from two main sources: TESS and Gaia DR3. TESS gives high-cadence photometry with a 2-minute time resolution, which helps pick up short eclipses and makes it easier to see details in the light curve like the eclipse depth, secondary eclipses, and orbital periods (Guerrero et al. 2021). Some systems also showed signs of orbital eccentricity, which affects the timing and shape of the eclipses. // Gaia DR3 was also used as it provides accurate astrometric and photometric data. With parallax, distances could be worked out, and when combined with the star’s apparent magnitude, gave us the absolute luminosity. Gaia also gives colour data through the G, BP, and RP filters, which helps estimate effective temperatures and colour indices (Morello et al. 2022). These are useful for working out limb darkening and comparing to theoretical models.

For modelling the light curves, we used the Lightkurve Python package to clean and phase-fold the TESS data. This helped remove any trends or outliers that could throw off the results. Then we used jktebop to model the systems and get key values like the fractional radii, inclination, surface brightness ratio, and for eccentric binaries, the $e \cos \omega$ and $e \sin \omega$ values (Southworth 2013). To improve the fits, we added limb darkening laws which are more important near the start and end of the eclipse (Southworth 2013).

To check if the modelled parameters were reliable, we used Monte Carlo simulations and residual permutations to estimate the uncertainties. This step is important because if the error bars are too big, for things like inclination or radius, the binary might not be good enough for PLATO, even if the light curve looks fine.

EBs aren’t just useful for calibration either. They can show things like tidal circularisation, angular momentum changes, and magnetic activity, which are all linked to how stars evolve. If a system has a noticeable eccentricity, it can also give clues about its past and how it formed. These EBs are useful for improving models in stellar population studies and planet formation research.

By using selected systems, combining TESS and Gaia data, and applying careful modelling, this project supports PLATO’s aim to do accurate exoplanet and stellar science. The next section will explain how we went about identifying and analysing the EBs in the LOPS2 field.

2 Methodology

2.1 Project aims and Target selection

The objective of the group project was to characterise detached EBs with high-precision photometry and astrometric data to identify suitable EB systems that could serve as a benchmark for the PLATO mission. This relies on understanding stellar properties to find exoplanets. The project stars selected will provide PLATO with a more refined selection of stars to target and help it cross-check and refine the stellar evolution model.

The focus when selecting stars from the published TESS eclipsing binary catalogue such as Prša et al. (2022), filtering with low morphology (< 0.6), ensuring the system is detached, minimising total deformation, and finding EBs with the least ellipsoidal effect and a suitable period to avoid eccentric stars. The selected stars analysed go through the process of cleaning the data, creating a light curve, and fitting the model using JKTEBOP (Southworth 2011). From the (~ 100 EBs), we reduced it to a few per person for deep analysis, and then we proceeded to find which of these stars fall within the P1 and P5 categories for PLATO’s target selection (Rauer et al. 2024).

2.2 Workflow for PLATO Benchmark Selection

The selected EBs from the TESS catalogue were processed in a Jupyter Notebook using a five-step workflow.

2.2.1 TESS and GAIA Data Processing and Model Fitting

The first step of this workflow focused on retrieving TESS data and cleaning the data via the TIC number, for example, TIC 349480507. Only data from the SPOC pipeline were used, with a 120-second cadence, and specific sectors were selected to improve phase coverage. The light curves were then normalised and converted to Barycentric TESS Julian Date (BTJD). Each sector was plotted and stitched together to identify any outliers. These outliers were then removed using manual data masking, aided by an interactive plot tool (example figure). This removal was critical because it isolated the eclipse features to ensure instrumental noise or flux variation did not interfere with the model fitting (figure-phase Lc).

Once outliers were removed, the clean and stitched light curve was carefully reviewed, and the orbital period between the eclipses was estimated with an uncertainty of 0.1. Finally, the phase-folded light curve of the EB was plotted. This was examined for any ellipsoidal variations; if above the acceptable threshold, the system would be excluded as

a PLATO candidate. The example star TIC 349480507 remained within the acceptable range. To generate the input files required for JKTEBOP, a Python script was used to convert the cleaned, phase-folded light curve into the correct format (`.dat file`). This ensured that JKTEBOP could accurately interpret the data for light curve modelling in the next step.

The same output was needed for GAIA data to model through JKTEBOP. There was a change in the formatting of the FITS file, but the required adjustment was needed to process the relevant information. The GAIA data was retrieved by cross-matching the TIC number and downloading the FITS file from the GAIA archive. Once that was done, the relevant FITS files were downloaded; these were then read and parsed using Astropy tools to extract flux and time data. The light curves were cleaned separately for each of the three photometric bands (G, BP, RP) to remove any outliers or poor data points. The data for each band was then normalised and phase-folded individually. The final processed outputs were saved in three separate `.dat` files corresponding to each band, ensuring compatibility with JKTEBOP. Allowing TESS and GAIA datasets to be modelled consistently using the same approach.

2.2.2 Modeling with JKTEBOP

JKTEBOP was used to model the phase-folded TESS light curves; this is done as JKTEBOP fits a binary system model based on geometric parameters, adjusting parameters based on the advice of the Lecturer, but the key is to minimise χ^2 between the observed and model light curves. Parameters such as orbital period and the reference time of primary minimum were values initially obtained from the TESS master notebook python scrip that processed the data, while the limb darkening parameters and gravity darkening coefficient were based on the temperatures of the EB (table provided by Lecturer). However, parameters like orbital inclination i , the ratio of radii $k = R_2/R_1$, the sum of fractional radii $(R_1 + R_2)/a$, the surface brightness ratio, and if applicable, the eccentricity terms $e \cos \omega$ and $e \sin \omega$, were all iteratively adjusted to align the light curve and the model fit.

A JKTEBOP `.in.3` input file was created to define these system parameters and specify which to adjust. Initial guesses were made using the shape and depth of the eclipses in the phase folded light curve; for most systems, eccentricity is assumed to be zero unless the orbital period and separation are large. Then, it will be adjusted to see if it could help with the alignment. For example, TIC 349480507 showed a secondary eclipse at a phase of 0.5, so no adjustment was needed for eccentricity, and other aspects such as scale factor, radii sum, surface brightness, third light, reflection A and B, and periods and

reference time were changed for better alignment. The process generated several output files, including the parameter file (`.out.3`), light curve fit file (`lcfit.out.3`), and the model fit file (`model.out.3`). The key was to repeatedly tweak the parameters until the model closely aligned with the observed light curve.

After completing the TESS light curve modelling, the next step was to apply the same process for the GAIA data using the G, BP, and RP photometric bands. Initially, using the respective band output files, the G band used `green.gaia.in.3`, BP used `blue.gaia.in.3`, and RP used `red.gaia.in.3`. These input files in the `.in.3` used the same parameters adjusted in the TESS `.in.3` file, such as period reference and reference time, and other parameters adjusted initially in TESS.

However, some parameters, such as surface brightness and scale factor, were adjusted through trial and error; these two parameters were fine-tuned to have the best impact when fitting the light curve and aligned model. One important parameter that needed to be changed was limb darkening, which was refined using individual research from group member Dan Mattock, who found the appropriate coefficients based on the stars' temperatures.

We over plotted the model fits from GAIA and TESS using Python. The magnitude values from each `model['Magnitude']` column were normalised, and all bands were plotted together for direct comparison. We included residuals (O–C) beneath the light curve plot to visualise the difference between the model and data. The aim was to flatten the residuals as much as possible. Any consistent mismatch that couldn't be resolved through parameter tweaking often pointed to slight inaccuracies in limb darkening.

2.2.3 Uncertainty Estimation

Two techniques for error estimation are the Monte Carlo simulations (Task 8) and residual-permutation analysis (Task 9), both of which are built into JKTEBOP (Southworth 2011). Task 8 was implemented for this study as it gave us the most accurate errors due to low noise in the TESS light curve.

2.2.4 Colour-Magnitude Diagram Creation

Using Gaia photometric data, a Colour-Magnitude Diagram (CMD) was created and compared to MIST isochrones. This helped check whether both stars are consistent with main sequence expectations and physically realistic as PLATO benchmarks.

The first part of the process involved collecting key values from the Gaia archive. From the Astrometry tab, we used the parallax and its uncertainty—for TIC 349480507,

this was 3.3461 ± 0.0115 mas. We pulled extinction in the G band (A_G) and the colour excess $E(BP - RP)$ from the Astrophysical Parameters tab, including their upper and lower limits. These were needed to correct the observed magnitudes for interstellar effects.

We then added the scale factor and flux ratio values for the G, BP, and RP bands: these came from the JKTEBOP model `param.out.8` (Task 8 output) output files. They helped split the total observed light between the two stars so we could place each one on the CMD individually. Errors were properly carried through using propagation, so uncertainties in the plot reflected the real data.

Using these parameters, we plot the corrected magnitudes and colours on the CMD alongside a range of isochrones (0.5 to 5 Gyr) to estimate mass and age. Shown in Figure 10, this confirmed whether the stars follow a realistic evolutionary path, which is important in deciding whether the EB is worth keeping as a PLATO benchmark.

2.2.5 Benchmark Evaluation

Building on the CMD analysis, the final step in the workflow was to evaluate whether each system could serve as a PLATO benchmark. To be a suitable candidate, both components of the binary system must lie within the visual magnitude threshold set for PLATO's P1 ($V \leq 11$ mag) and P2 ($V \leq 13$ mag) samples.

However, it was not just about brightness. Visually, both stars had to fall within the central, well-populated region of the CMD, where main sequence stars lie, because PLATO prioritises solar-like, unevolved stars for asteroseismic and exoplanet studies. The system would not be considered a benchmark either component was significantly off the main sequence.

To assess this, the individual visual magnitudes of each component were calculated using flux ratios obtained from the JKTEBOP `param.out.8` files and the formula derived in the working shown (see Figure ??). The total system magnitude V_b was retrieved from the Gaia archive, and then converted to flux via:

$$F_b = 10^{-0.4V_b} \quad (1)$$

Given the flux ratio $\ell = F_2/F_1$, we split the total flux into component fluxes, and calculated individual magnitudes V_1 and V_2 :

$$V_1 = V_b + 2.5 \log_{10}(1 + \ell) \quad (2)$$

$$V_2 = V_b + 2.5 \log_{10} \left(1 + \frac{1}{\ell} \right) \quad (3)$$

This helped determine whether each component met the visual magnitude criteria separately and appeared on the main sequence. EBS passing this benchmark check were retained as a priority for the PLATO mission.

3 Results

This section presents the results obtained from the analysis of the two primary EBs assigned and thought to have minimum ellipsoidal effect: TIC 349480507 and TIC 349059354. These were modelled using the workflow outlined in Section 2, including data processing, light curve modelling, CMD placement, and benchmark evaluation. The final and third system was initially selected, but due to issues retrieving usable GAIA data, it was excluded (more in section 4). The results for each star are presented individually, followed by a summary of the group’s benchmark systems, where all analysed EBs are compared and evaluated as PLATO benchmark candidates.

3.1 TESS Phase-Folded Light Curves

Figure 3 and Figure 2 show the final TESS phase-folded light curves for TIC 349480507 and TIC 349059354, respectively. These light curves were processed as outlined in Section 2.2.1, using the SPOC pipeline data with a 120-second cadence. Each sector was normalised, stitched, and cleaned via manual masking to remove outliers before being phase-folded using the orbital period estimated from the eclipse timings.

The eclipse depths and overall symmetry of the curves allowed initial estimates of key system parameters such as inclination and radius ratio. This provided a good starting point for model fitting in JKTEBOP.

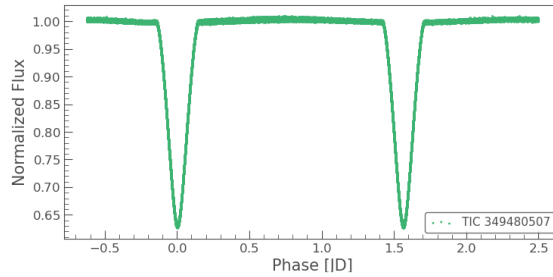


Figure 1: TESS phase-folded light curve for TIC 349480507. Primary secondary components are similar in brightness and are similar size. the light curve has ellipsoidal effect but does not exceed the groups threshold for acceptability, which TIC 349480507 is the upper limit in the detached sample selection.

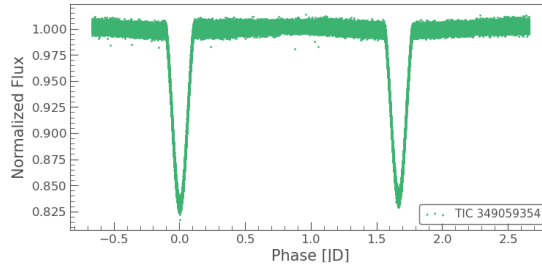


Figure 2: TESS phase-folded light curve for TIC 349059354. The primary and secondary eclipses are clearly visible and have similar depths, suggesting the two components have comparable brightness and possible similar size. The data was cleaned and stitched from multiple sectors and show minimal ellipsoidal effect , but still above the threshold.

Only TIC 349059354 was successfully phase-folded using both TESS and Gaia data, as this allowed visual alignment to check if the GAIA bands had data on top of the TESS light curve; this was done to make sure $T_{\min}$ adjustment was not needed. TIC 349480507 was excluded as the GAIA FITS file could not be opened and over plotted; however, the

.dat file was generated for all bands.

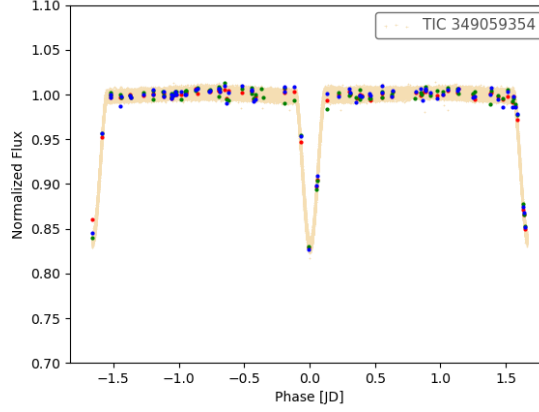


Figure 3: Phase-folded light curve of TIC 349059354 showing over plotted Gaia data (G, BP, RP bands) on top of the TESS light curve. The Gaia points align well, this means that $T_{\min}$ was not recalculated, as the fit was sufficient.

3.2 Light curve modelling - JKTEBOP

TIC 349059354 was modelled using JKTEBOP, the TESS light curve was first fitted, then the Gaia G, BP and RP bands were modelled by adjusting only the surface brightness and scale factors. All three bands showed good fits with low residuals, and the final overplot (Figure 6).

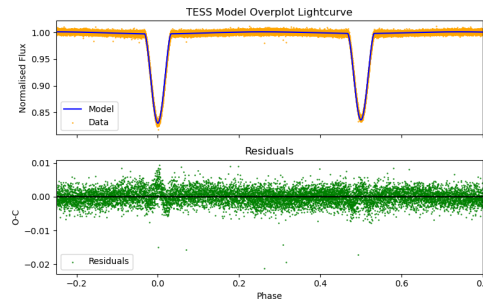


Figure 4: TESS light curve model fit for TIC 349059354 from JKTEBOP. Top panel shows model overplot (blue) and observed data (orange). Bottom panel shows residuals (green), which are centred around zero, confirming good agreement.

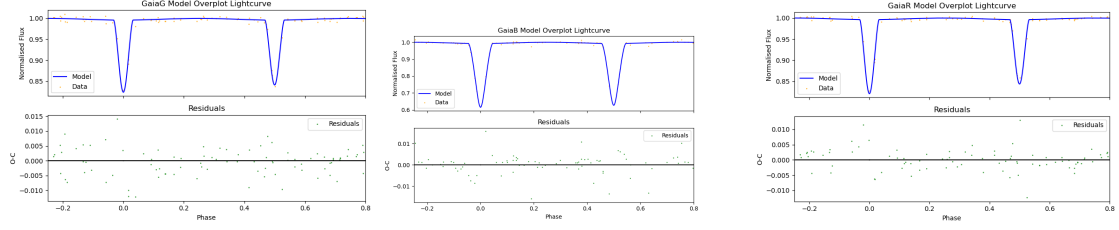


Figure 5: Model fits for Gaia G (left), BP (middle), and RP (right) bands for TIC 349480507. The fit is acceptable overall but slightly mismatched at the primary eclipse.

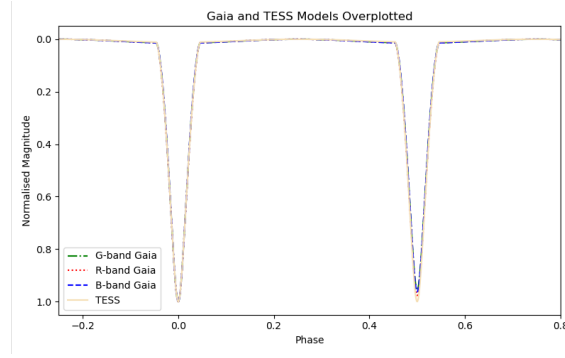


Figure 6: Overplot of TESS and Gaia G, BP, and RP band data for TIC 349059354, consistent eclipse alignment across bands confirms that the same parameters describe all datasets well.

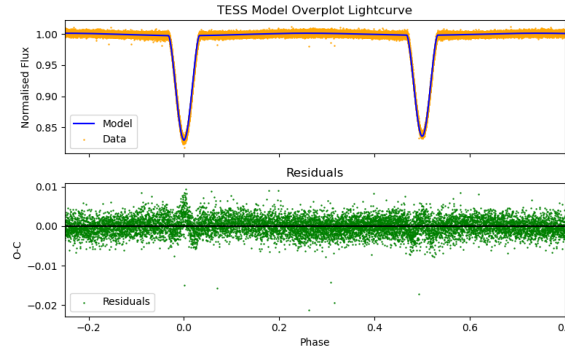


Figure 7: TESS model fit for TIC 349480507 with residuals. Parameters were adjusted to flatten residuals, which worked well for the secondary eclipse but not the primary.

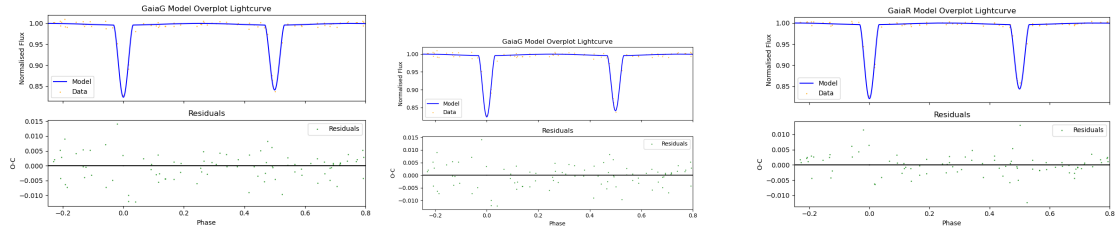


Figure 8: Model fits for Gaia G (left), BP (middle), and RP (right) bands for TIC 349480507. The fit is acceptable overall but slightly mismatched at the primary eclipse.

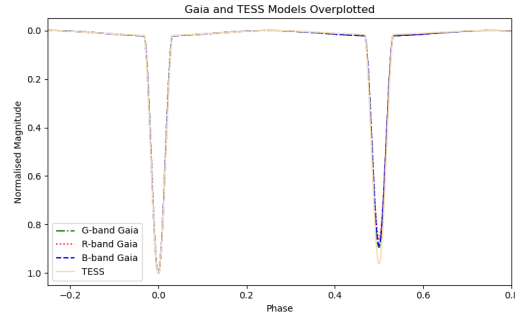


Figure 9: Overplot of model fits for TESS and Gaia bands for TIC 349480507. All fits align well, confirming parameter consistency.

3.3 Colour-Magnitude Diagram and Benchmark Selection

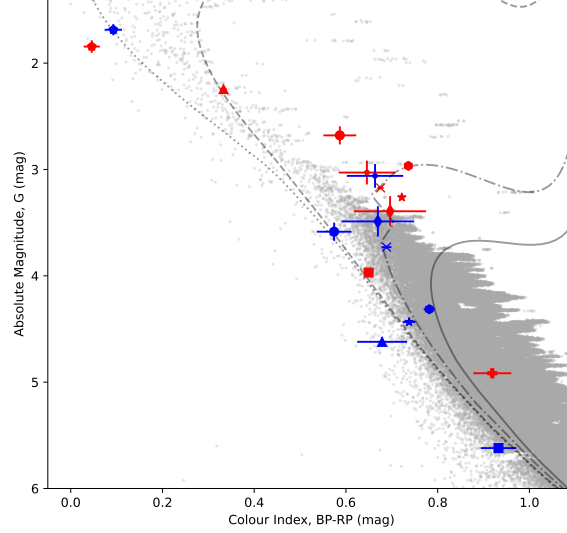


Figure 10: Colour-Magnitude Diagram (CMD) showing the primary (red) and secondary (blue) components of each binary system analysed in the group. Isochrones from Dotter (2016) and Paxton et al. (2018) span stellar ages from 0.5 to 10 Gyr (left to right), overplotted on a background of Gaia main-sequence stars (Gaia Collaboration et al. 2018). Systems are identified using different labels: TIC279087522 (pentagon), TIC30122338 (circle), TIC7695666 (cross), TIC66602813 (square), TIC63579446 (star), TIC201497357 (triangle), TIC349480507 (dot), TIC349059354 (diamond), TIC78568736 (plus; secondary lies outside plot range), and TIC80556181 (hexagon). Systems positioned near the centre of the main sequence are more likely to be PLATO benchmarks, as they may be solar-like. Systems in the upper-left region, where primary components appear fainter than secondary components, were likely affected by light curve inversion or incorrect flux assignment.

Table 1: Visual band magnitudes (V) for each binary system in the group, including the associated uncertainty (V_{err}) and the derived visual magnitudes for the primary (V_1) and secondary (V_2) components. These values were obtained using the formulas in 2.2.5

Source Symbol	TESS ID	V	V_{err}	V_1	V_2
square	TIC66602813	9.195	0.002	9.280853	9.9412491409
circle	TIC30122338	9.991	0.005	10.147565	10.5100219767
cross	TIC7695666	10.675	0.008	10.878776	11.1015434379
pentagon	TIC279087522	9.875	0.005	10.208781	10.1455764750
star	TIC63579446	11.521	0.013	11.648008	12.1169102141
triangle	TIC201497357	8.400	0.030	8.446116	9.3967881620
point	TIC349480507	9.959	0.004	10.253799	10.2663519531
diamond	TIC349059354	11.347	0.011	11.629358	11.6675411543
plus sign	TIC78568736	9.825	0.003	9.863751	10.8937307601
hexagon	TIC80556181	10.746	0.074	10.856230	11.3954345179

4 Discussions

4.1 Eccentricity in Eclipsing Binaries

One of the fascinating parts of this study is the orbital eccentricity, which was linked to the period and the separation of the components. How large or small must the period be before eccentricity starts affecting data and making light curve analysis difficult? Tidal interaction theory suggests that binaries have a threshold at which they are circular, and outside this threshold, they become eccentric (Justesen & Albrecht 2021).

Justesen & Albrecht (2021)) found that systems with periods below roughly 10 days are typically circularised, especially among F- and G-type stars, whereas longer-period systems retain significant eccentricity. This was confirmed as most of the binaries studied were between 3–8 days, and these binaries show low to no eccentric behaviour. Then, when looking at two EB systems that had longer-period binaries, TIC7695666 and TIC78568736, the binary system had periods of 17.6 and 8.2 days respectively, confirming Justesen & Albrecht (2021) as they showed non-zero eccentricities when modelled in JKTEBOP.

On the other hand, there were EBS that had periods shorter than 3 days, which also required adjustment of JKTEBOP to accurately fit the light curve and flatten the residuals.

TIC 307084982 was one of the EBS analysed; this star had the data for TESS processing and cleaning, but did not have the GAIA epoch photometry available, limiting the analysis. However, the overplot of the model and the light curve fit was analysed and adjusted using its `.in.3` file, the scale factor and other standard parameters, which we have been changing, and now it includes eccentricity. A clean and flat residual plot was the result of manually adjusting eccentricity and through trial and error, as shown in Figure 11. While it was not a complete system analysis using the GAIA bands, it helps to understand how eccentricity with extensive periods, for example, TIC 307084982, had a 19.2 days orbital period, so it indicated before any literature review that large periods increase the probability of an EB having eccentric behaviour.

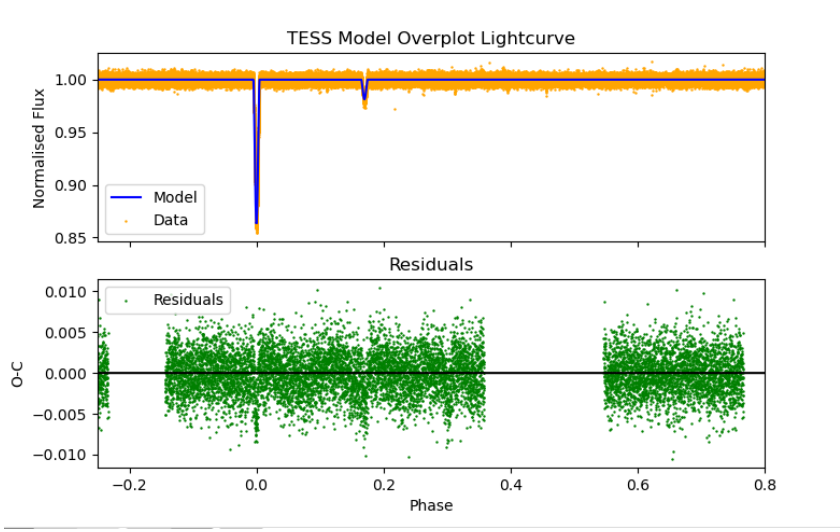


Figure 11: TESS phase-folded light curve of TIC 307084982 with the JKTEBOP model (blue) overplotted on the data (orange, top panel) and the corresponding residuals (green, bottom panel). The flat residuals demonstrate that including an eccentricity in the model yields an excellent fit, even in the absence of data missing.

These findings align well with the recent study by Siqueira et al. (2024), who found that the circularisation period for eclipsing binaries is closer to 5.76 days, lower than earlier theoretical models predicted. Our dataset partially supports this, as we observed that systems with periods above 8 days often showed measurable eccentricity. When averaging the orbital periods of binaries in our study that appeared to have circular orbits, the value was close to 5.76 days. This agreement suggests that mid-range periods are typically circularised. In contrast, systems with much shorter or significantly longer periods are more likely to retain or develop eccentric orbits.

In the context of PLATO’s mission, avoiding eccentric binaries is essential as they present challenges that will take too long to study and align. They may not be good candidates for many reasons, such as that they are likely young stars or even radiative envelopes if they retain eccentricity. For example, Siqueira et al. (2024) report a critical circularisation period of approximately 5.76 days specifically for stars with radiative envelopes, significantly lower than old model estimates. The long-period systems found to show measurable eccentricity fit this updated picture well.

4.2 discussion

This project aimed to identify and analyse detached EBS in the LOPS2 field that could become benchmarks for the PLATO mission. These benchmark stars were expected to have solar-like masses ($\sim 1M_{\odot}$), well-determined parameters, and apparent magnitudes that fall within either the strict $P1$ criteria ($V < 11$ mag) or the dimmer and broader $P5$ detection range. The study modelled ten full systems by the group, and additionally, an analysis was done on TIC 307084982; the results strongly indicate that the aim was successful.

To determine key parameters such as orbital period, eccentricity, inclination, and component masses, TESS light curves and GAIA DR3 data were used. A final CMD (Figure 10), was produced and confined to the EBS studied, which were mostly in the main sequence with some outliers. Six of the ten group systems all satisfied the $P1$ sample specification. The EBS were TIC 66602813, TIC 30122338, TIC 7695666, TIC 349480507, TIC 78568736, and TIC 80556181. These targets also have near-solar-like masses, making them excellent candidates for the PLATO mission. The periods of the EBS were within the range of non-eccentric orbits.

All ten EBs met the $P5$ sample specification ($V < 13$ mag). However, two systems, TIC 279087522 and TIC 201497357, while bright enough to be detected, have primary masses of approximately $2M_{\odot}$ and $1.8M_{\odot}$, respectively, which place them well outside the solar-like mass range. As such, these two systems are considered lower priority for PLATO benchmark selection.

There were some EBs such as IC 78568736, which had their primary and secondary components the other way around (Top left of Figure 10), this was due its secondary component that contributes only $\sim 10\%$ of the total flux and therefore the lack of GAIA coverage during eclipse, limited the precision of the secondary parameters.

In conclusion, this project has successfully identified several eclipsing binaries (EBs) that PLATO can prioritise as benchmark systems in its future mission. As a group, we've highlighted multiple high-quality candidates and demonstrated how detailed modelling, especially in systems with eccentric behaviour, enhances our understanding of each EB's suitability. Future improvements should focus on ensuring that all stars have complete `.dat` files prepared early in the workflow. During our analysis, Gaia updated the formatting of their band data, which required extra time spent on Python scripting just to process the files correctly. Although this was eventually resolved, addressing such issues at the start would save time and ensure smoother progress in future studies.

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